

2.2.3 The mammalian gas exchange system

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) (i) the relationships between cells, tissues and organs, with reference to the mammalian gas exchange system	To include the appearance and the histology of squamous epithelial cells in the alveoli, ciliated epithelial tissue, smooth muscle, cartilage and elastic fibres.
(ii) observations of tissues of the gas exchange system using microscopy	PAG1 HSW4, HSW5, HSW6
(b) the process of gas exchange in the alveoli	To include the roles of ventilation, epithelial tissue, smooth muscle, cartilage, elastic fibres, blood capillaries and surfactant in the establishment and maintenance of concentration gradients.
(c) the parameters affecting pulmonary ventilation	To include consideration of tidal volume, breathing rate, vital capacity, residual volume, PEFR and FEV ₁ (no detail of the use of a spirometer is required). PAG10 M0.1, M2.2, M2.3, M2.4
(d) how expired air resuscitation can be carried out on adults, children and babies in cases of respiratory arrest.	To include reference to both manual and electronic methods. HSW3, HSW4, HSW5, HSW6